

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer

1. Error Identification

The neural network produces a **training Mean Squared Error (MSE) of 0.85** and a **testing MSE of 5.20**. The large difference between these values indicates that the model is **overfitting**. It has learned the training data very well but is unable to perform equally well on unseen test data. Possible causes include an overly complex neural network with too many layers or neurons, a small or unrepresentative training dataset, noisy or irrelevant input features, excessive training epochs, and the absence of regularization techniques such as dropout or L2 regularization.

2. Error Analysis & Reasoning

Overfitting occurs when the neural network memorizes the patterns and noise present in the training data instead of learning the underlying relationship between the input features and student scores. As a result, the model achieves a very low training MSE because it fits the training samples closely. However, when new test data is provided, the learned patterns do not generalize well, causing the testing MSE to increase significantly. This large gap between training and testing performance indicates poor generalization and reduced predictive capability. The issue may also arise if the training and testing datasets have different data distributions or if the training data is insufficient to represent all possible student performance patterns.

3. Correction Strategy

Several corrective measures can reduce overfitting and improve the model's performance:

- **Reduce model complexity** by decreasing the number of hidden layers or neurons so that the network learns only essential patterns.
- **Apply regularization techniques** such as L1 or L2 regularization to penalize overly large weights and prevent the model from fitting noise.
- **Use dropout layers**, which randomly deactivate neurons during training, encouraging the network to learn more robust features.
- **Implement early stopping** by monitoring the validation loss and stopping training before overfitting begins.
- **Increase the size and diversity of the training dataset** or apply data augmentation where applicable so the model learns more generalized patterns.
- **Perform feature selection** to remove noisy or irrelevant features that negatively affect learning.
- **Use cross-validation** to evaluate the model on multiple data splits and ensure consistent performance.
- **Tune hyperparameters**, including learning rate, batch size, number of epochs, and optimizer settings, to achieve better generalization.

These methods help the neural network learn meaningful relationships rather than memorizing the training data.

4. Interpretation of Results

After applying the corrective measures, the testing MSE is expected to decrease significantly and become much closer to the training MSE. A smaller gap between training and testing errors indicates that the model has improved its ability to generalize to unseen data. The neural network will provide more accurate and reliable predictions of student scores, reduce prediction errors, and achieve better overall performance. This demonstrates that the model has learned the underlying patterns in the data rather than simply memorizing the training examples, making it more suitable for real-world applications.

Code of Conduct:

*I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided